

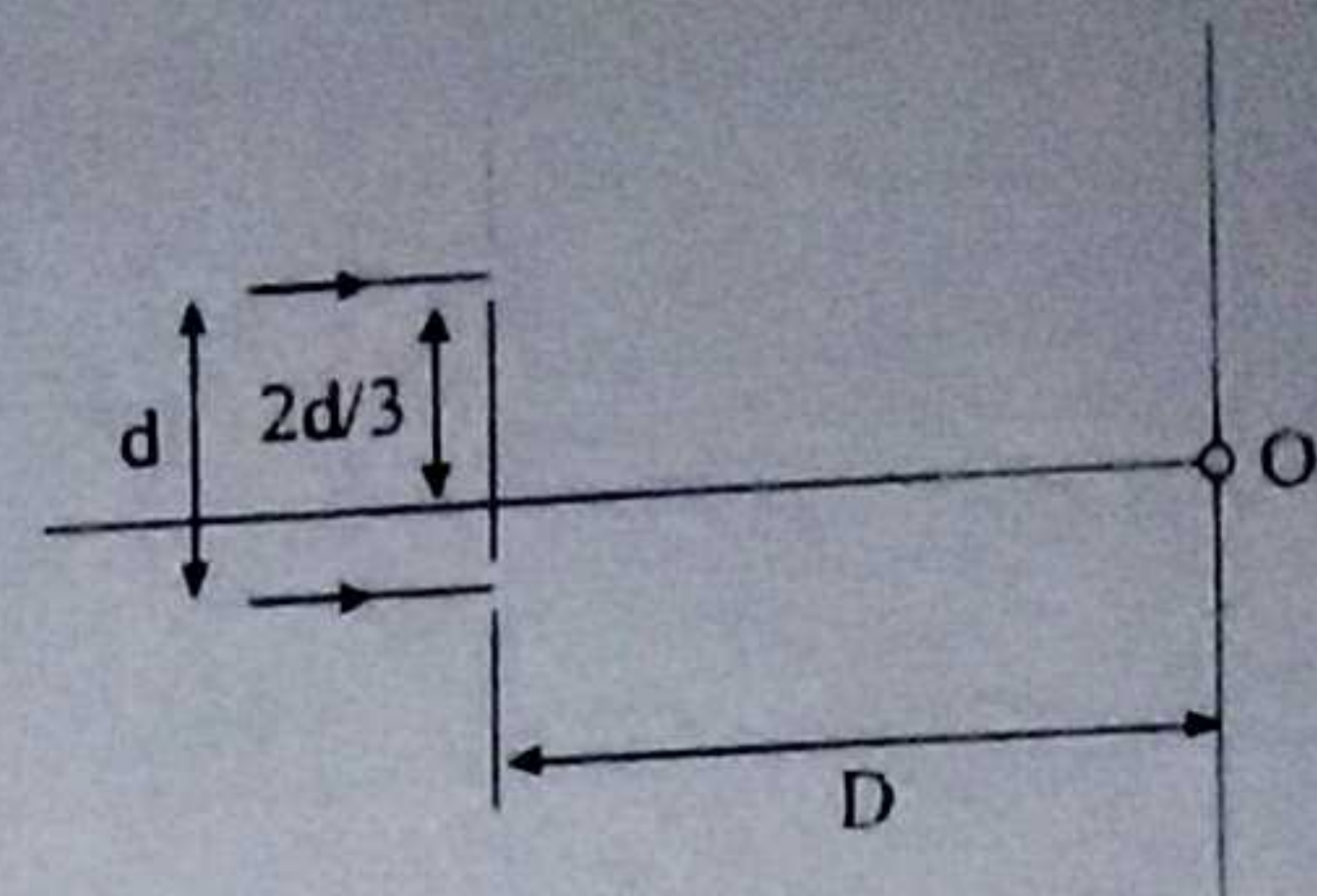
Wave Optics

Type of CPP - Objective & Subjective

Level - MODERATE

- In Young's interference experiment, if I_0 be the intensity at the central bright fringe, β be the fringe width then the intensity as a function of distance x from the central bright fringe is :
 (A) $I = I_0 \cos^2\left(\frac{x}{\beta}\right)$ (B) $I = I_0 \cos^2\left(\frac{\beta}{x}\right)$ (C) $I = I_0 \cos^2\left(\frac{\pi\beta}{x}\right)$ (D) $I = I_0 \cos^2\left(\frac{\pi x}{\beta}\right)$
 - A parallel beam of monochromatic light is used in a Young's double slit experiment. The slits are separated by a distance d and the screen is placed parallel to the plane of the slits. If the incident beam makes an angle $\theta = \sin^{-1}\left(\frac{\lambda}{2d}\right)$ with the normal to the plane of the slits, find the intensity at the geometrical centre P_0 .
 - In Young's double slit experiment, if distance between the slit is doubled, then to keep fringe width unchanged :
 (A) wavelength should be made double
 (B) distance between screen and slit should be made double
 (C) both should remain constant (D) none of these
 - Two coherent light sources each of wavelength λ are separated by a distance 3λ . The number of minima formed on line AB which runs from $-\infty$ to $+\infty$ is :
 (A) 2 (B) 4 (C) 6 (D) 8
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- Two identical sources each of intensity I_0 have a separated $d = \frac{\lambda}{8}$, where λ is the wavelength of the wave emitted by either source. The phase difference of the sources is $\frac{\pi}{4}$. The intensity distribution $I(\theta)$ in the radiation field as a function of θ , which specifies the direction from the sources to the distance observation point P is given by :
 (A) $I(\theta) = I_0 \cos^2\theta$ (B) $I(\theta) = \frac{I_0}{4} \cos^2\left(\frac{\pi\theta}{8}\right)$
 (C) $I(\theta) = 4I_0 \cos^2\left[\frac{\pi}{8}(\sin\theta + 1)\right]$ (D) $I(\theta) = I_0 \sin^2\theta$
 - In Young's double slit experiment, interference pattern is found to have an intensity ratio between bright and dark fringes as 9. Then :
 (A) Intensities on the screen due to two slits are 5 units and 4 units respectively
 (B) amplitude ratio is 2 (C) amplitude ratio is 9
 (D) intensities at the screen due to two slits are 4 units and 1 unit respectively
 - The Young's double slit experiment is performed with blue and with green light of wavelengths $4360 \text{ \AA}$ and $5460 \text{ \AA}$ respectively. If X is the distance of 4th maximum from the central one, then :
 (A) $X(\text{blue}) = X(\text{green})$ (B) $X(\text{blue}) > X(\text{green})$ (C) $X(\text{blue}) < X(\text{green})$ (D) $\frac{X(\text{blue})}{X(\text{green})} = \frac{5460}{4360}$

8. In the figure shown in a parallel beam of white light is incident on the plane of the slits then the distance of the nearest white spot on the screen from O is :
 (A) 0
 (B) $d/2$
 (C) $d/3$
 (D) $d/6$

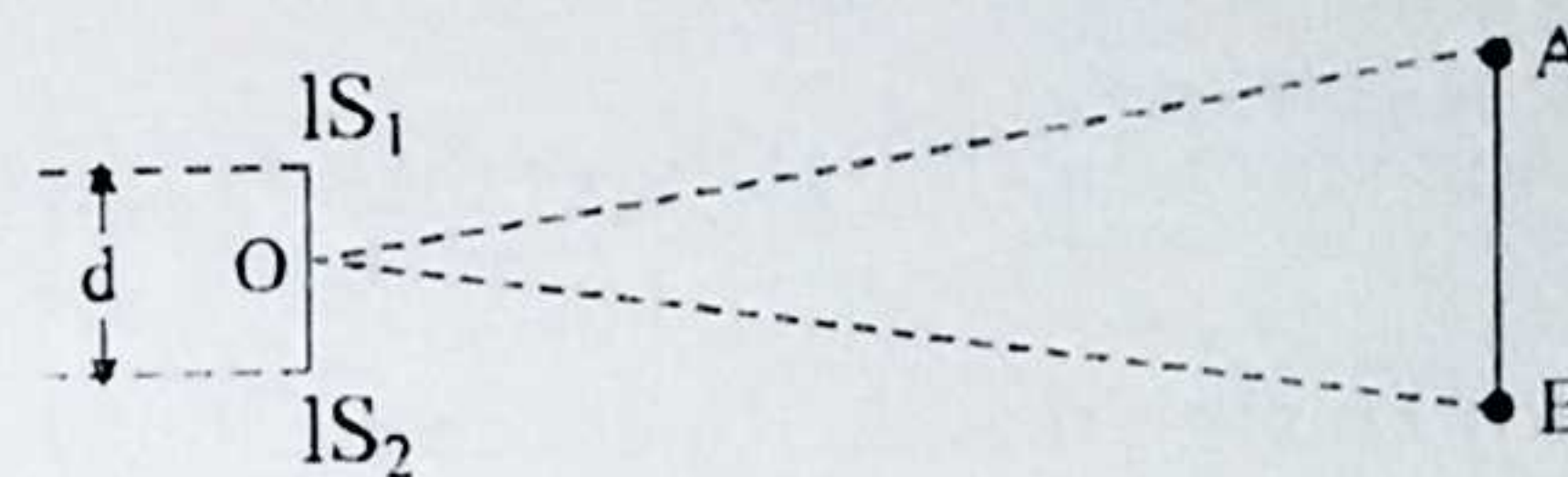


9. A beam of light of wavelength 600 nm from a distance source falls on a single slit 1 mm wide and the resulting diffraction pattern is observed on a screen 2 m away. The distance between the first dark fringes on either side of the central bright fringe is :
 (A) 1.2 cm (B) 1.2 mm (C) 2.4 cm (D) 2.4 mm

10. Find the thickness of a plate which will produce a change in optical path equal to half the wavelength λ of the light passing through it normally. The refractive index of the plate is μ .

11. Two slits are 0.3 mm apart. The interference fringes are formed on a screen 0.9 metre away by illuminating the slits by mono-chromatic light. If interference pattern the 2nd dark fringes is formed at 3 mm distance from a central fringe. then the wavelength of light will be :
 (A) 6000 Å (B) 6700 Å (C) 4000 Å (D) 5000 Å

12. Figure shows two coherent sources S_1, S_2 vibrating in same phase. AB is an irregular wire lying at a far distance from the sources S_1 and S_2 . Let $\frac{\lambda}{d} = 10^{-3}$ $\angle BOA = 0.12^\circ$. How many bright spots will be seen on the wire, including points A and B .



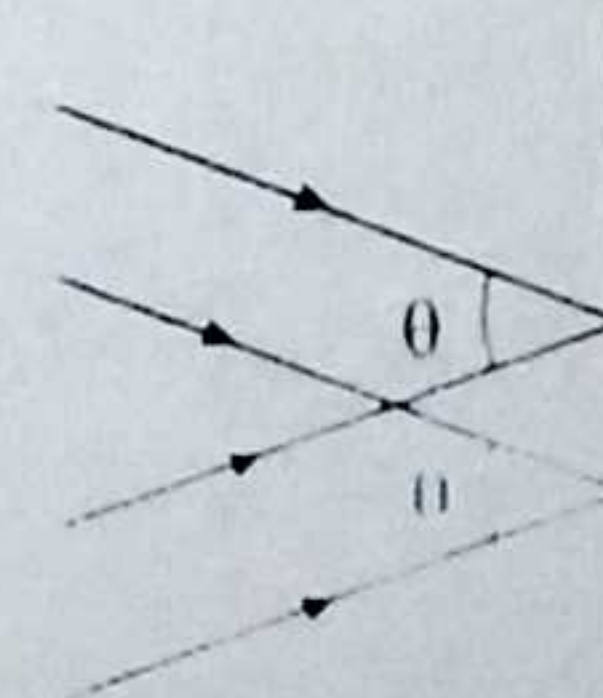
13. In a YDSE both slits produce equal intensities on the screen. A 100% transparent thin film is placed in front of one of the slits. Now the intensity of the geometrical centre of system on the screen becomes 75% of the previous intensity. The wavelength of the light is 6000 Å and $\mu_{\text{glass}} = 1.5$. The thickness of the film cannot be :
 (A) 0.2 μm (B) 1.0 μm (C) 1.4 μm (D) 1.6 μm

14. In a Young's double slit experiment, when one of the slits is covered with transparent of the slits is covered with transparent slab of thickness 3.6×10^{-3} cm, then the central fringe is displaced to the place of 30th bright fringe. If $\lambda = 6000$ Å, then the refractive index will be :
 (A) 1.2 (B) 1.5 (C) 2.4 (D) 3.0

15. In Young's double slit experiment, light of wavelength 5000 Å is used. If a thin transparent foil of thickness 2.5×10^{-3} cm is placed in the path of light from the slits, then the interference pattern is displaced by a distance equal to twenty fringe widths. The refractive index of the foil :
 (A) 2.3 (B) 1.95 (C) 1.6 (D) 1.4

16. Two parallel beams of light of wavelength λ inclined to each other at angle θ ($\ll 1$) are incident on a plane at near normal incidence. The fringe width will be :

- (A) $\frac{\lambda}{2\theta}$ (B) $\frac{2\lambda}{\theta}$
 (C) $\frac{\lambda}{\theta}$ (D) $2\lambda \sin \theta$



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ANSWER KEY

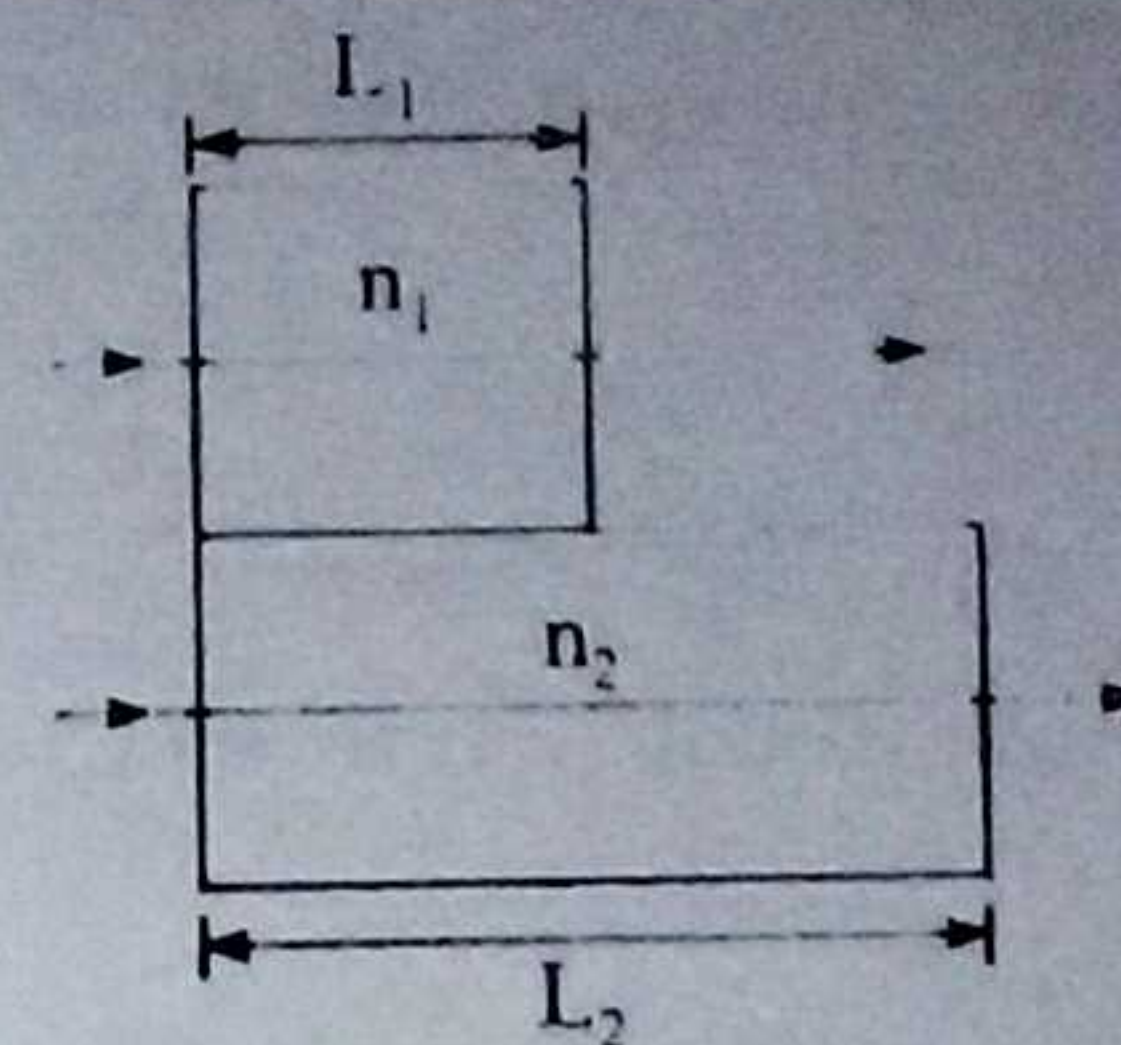
1. (D) 2. [Zero] 3. (A) (B) 4. (C) 5. (C) 6. (B) (D) 7. (C) 8. (D)
 9. (D) 10. $\left[\frac{\lambda}{2(\mu - 1)} \right]$ 11. (B) 12. [3] 13. (D) 14. (B) 15. (D) 16. (C)

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Type of CPP - Objective & Subjective

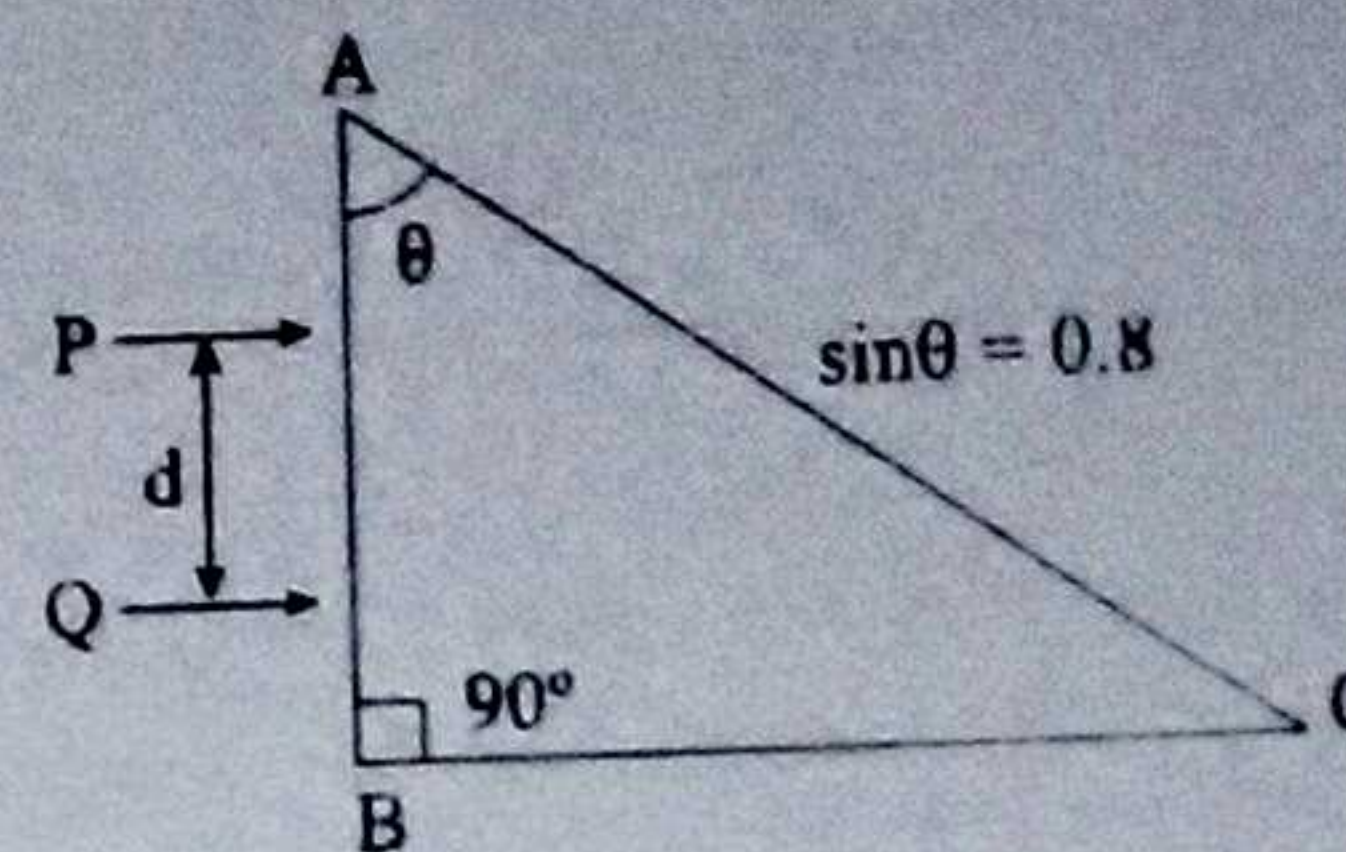
Level - MODERATE

1. Two waves of light in air have the same wavelength and are initially in phase. They then travel through plastic layers with thicknesses of $L_1 = 3.5 \mu\text{m}$ and $L_2 = 5.0 \mu\text{m}$ and indices of refraction $n_1 = 1.7$ and $n_2 = 1.25$ as shown in the figure. The rays later arrive at a common point. The longest wavelength of light for which constructive interference occurs at the point is :
- (A) $0.6 \mu\text{m}$ (B) $1.2 \mu\text{m}$
(C) $2.4 \mu\text{m}$ (D) $0.3 \mu\text{m}$
2. Find the angular separation between the consecutive bright fringes in a Young's double slits experiment with blue-green light of wavelength 500 nm . The separation between the slits is $2.0 \times 10^{-3} \text{ m}$:
- (A) 0.014° (B) 0.023° (C) 0.028° (D) 0.046°
3. If one of the slits of a standard Young's double slit experiment is covered by a thin parallel sided glass slab so that it transmits only one half the light intensity of the other, then :
- (A) The fringe pattern will get shifted towards the covered slit
(B) The fringe pattern will get shifted away from the covered slit
(C) The bright fringes will become less bright and the dark ones will become more bright
(D) The fringe width will remain unchanged
4. A parallel beam of light of wavelength 560 nm falls on a thin film of oil (refractive index = 1.4). What should be the minimum thickness of the film so that it strongly reflects the light ?
5. In a Young's double-slit experiment, let S_1 and S_2 be the two slits, and C be the centre of the screen. If $\angle S_1CS_2 = \theta$ and λ is the wavelength, the fringe width will be :
- (A) $\frac{\lambda}{\theta}$ (B) $\lambda\theta$ (C) $\frac{2\lambda}{\theta}$ (D) $\frac{\lambda}{2\theta}$
6. Find the thickness of a plate which will produce a change in optical path equal to half the wavelength λ of the light passing through it normally. The refractive index of the plate is μ :
- (A) $\frac{\lambda}{2(\mu-1)}$ (B) $\frac{\lambda}{\mu-1}$ (C) $\frac{2\lambda}{\mu-1}$ (D) $\frac{3\lambda}{2(\mu-1)}$
7. Laser light (of wavelength 630 nm) incident on a pair of slits produces an interference pattern in which the bright fringes are separated by 8.3 mm . A second light produces an interference pattern in which the bright fringes are separated by 7.6 mm . The wavelength of this second light is :
- (A) 688 nm (B) 577 nm (C) 600 nm (D) 630 nm
8. If instead of monochromatic light, white light is used in Young's double slit experiment, then :
- (A) a bright white fringe is formed at the centre of screen
(B) fringes of different colours are clearly observed only in the first order place
(C) the first order red fringes are relatively close to the screen than the first order violet fringes
(D) the first order violet fringes are relatively close to the centre of the screen than the first order red fringes



9. Two parallel beams of light P & Q (separation d) containing radiations of wavelengths $4000 \text{ \AA}$ & $5000 \text{ \AA}$ (which are mutually coherent in each wavelength separately) are incident normally on a prism as shown in figure. The refractive index of the prism as a function of wavelength is given by the relation.

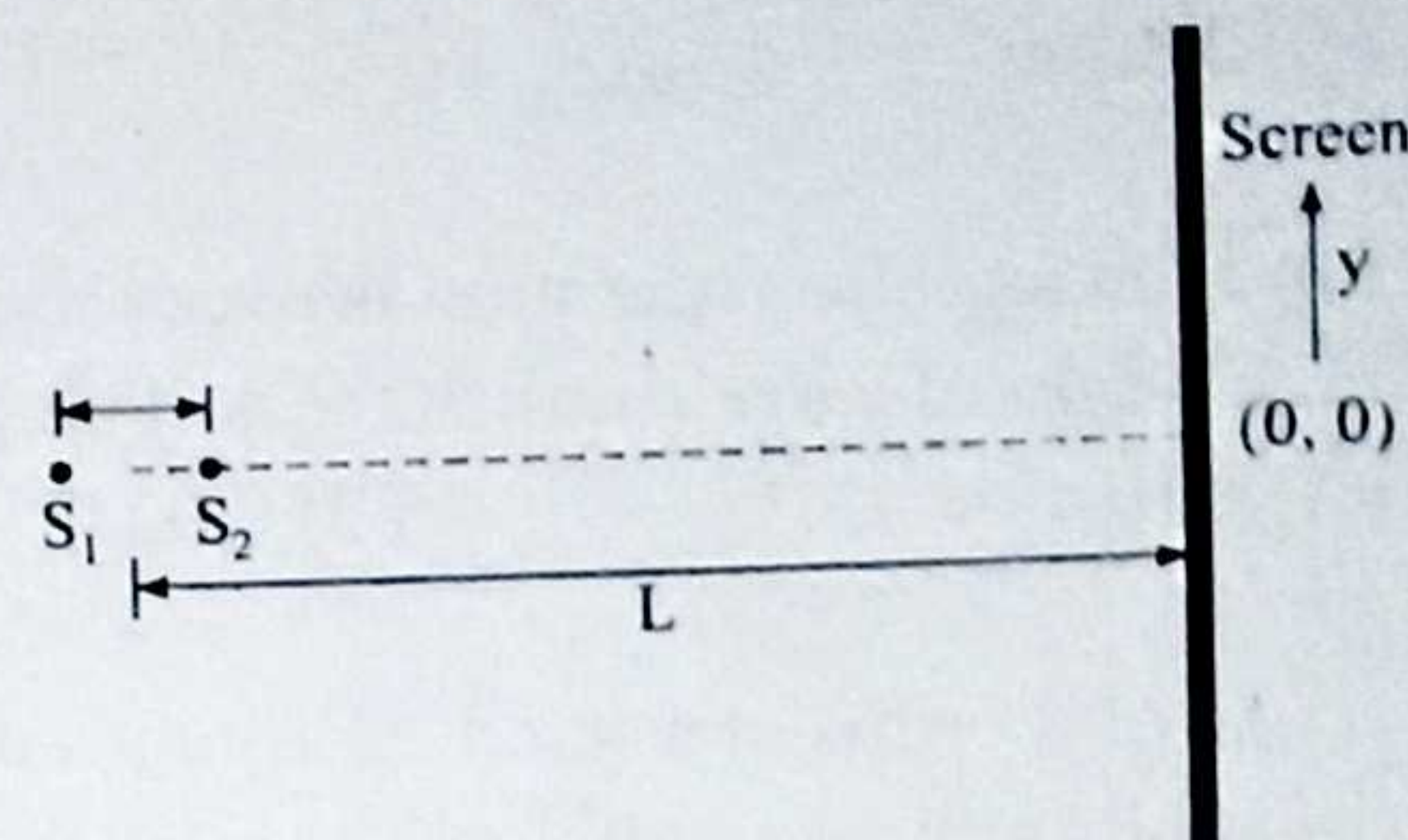
$\mu(\lambda) = 1.20 + \frac{b}{\lambda^2}$, where λ is in $\text{\AA}$ & b is a positive constant. The value of b is such that the condition for total reflection at the face AC is just satisfied for one wavelength & is not satisfied for the other. A convergent lens is used to bring these transmitted beams into focus. If the intensities of the upper & the lower beams immediately after transmission from the face AC are $4I$ & I respectively, find the resultant intensity, at the focus.



10. A soap film of thickness 0.0011 mm appears dark when seen by the reflected light of wavelength 580 nm . What is the index of refraction of the soap solution, if it is known to be between 1.2 and 1.5 ?
 (A) 1.21 (B) 1.32 (C) 1.4 (D) 1.48

11. A soap film of thickness 0.0011 mm appears dark when seen by the reflected light of wavelength 580 nm . What is the index of refraction of the soap solution, if it is known to be between 1.2 and 1.5 ?

12. The figure shows two point sources which emit light of wavelength λ in phase with each other and are at a distance $d = 5.5 \lambda$ apart along a line which is perpendicular to a screen at a distance L from the centre of the sources. Assume that d is much less than L . Which of the following statements is (are) correct :



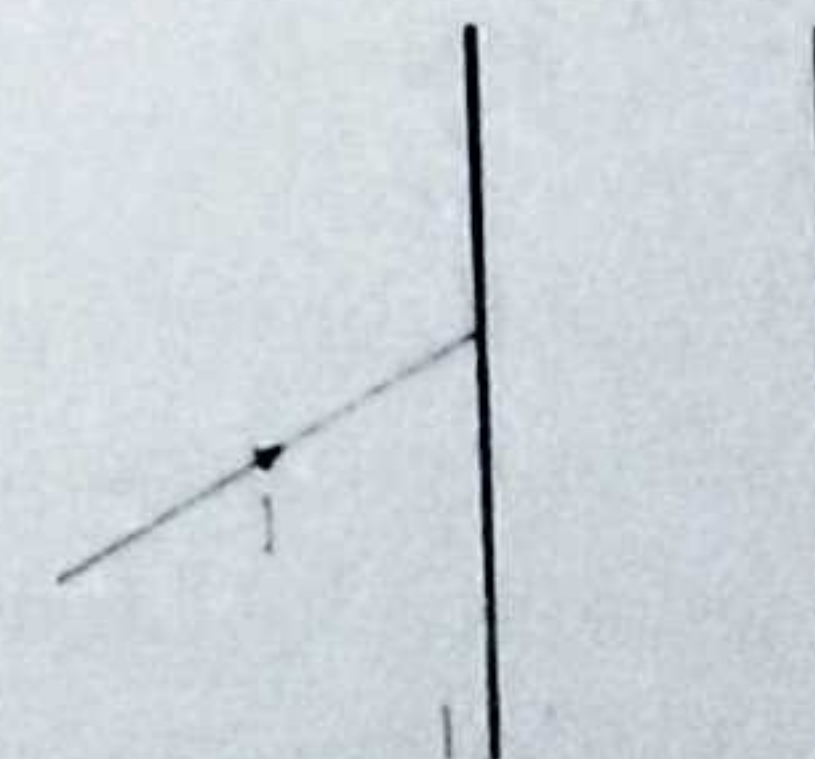
- (A) Only five bright fringes appear on the screen
 (B) Only six bright fringes appear on the screen
 (C) Point $y = 0$ corresponds to bright fringe
 (D) Point $y = 0$ corresponds to dark fringe

13. In the ideal double slit experiment, when a glass plate (refractive index 1.5) of thickness t is introduced in the path of one of the interfering beams (wavelength λ), the intensity at the position where the central maximum occurred previously remains unchanged. The minimum thickness of the glass plate is :
 (A) 2λ (B) $\frac{2\lambda}{3}$ (C) $\frac{\lambda}{3}$ (D) λ

14. In a Young's double slit experiment, $d = 1 \text{ mm}$, $\lambda = 6000 \text{ \AA}$ & $D = 1 \text{ m}$. The slits produce same intensity on the screen. The minimum distance between two points on the screen having 75% intensity of the maximum intensity is :
 (A) 0.45 mm (B) 0.40 mm (C) 0.30 mm (D) 0.20 mm

15. A flake of glass (refractive index 1.5) is placed over one of the openings of a double slit apparatus. The interference pattern displaces itself through seven successive maxima towards the side where the flake is placed. If wavelength of the diffracted light is $\lambda = 600 \text{ nm}$, then the thickness of the flake is :
 (A) 2100 nm (B) 4200 nm (C) 8400 nm (D) None of these

16. A narrow monochromatic beam of light of intensity I is incident on a glass plate as shown in figure. Another identical glass plate is kept close to the first one & parallel to it. Each glass plate reflects 25% of the light incident on it & transmits the remaining. Find the ratio of the minimum & the maximum intensities in the interference pattern formed by the two beams obtained after one reflection at each plate.



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ANSWER KEY

1. (B) 2. (A) 3. (A) (C) (D) 4. $[100 \text{ nm}]$ 5. (A) 6. (A) 7. (B)
 8. (A) (B) (D) 9. $[9 I]$ 10. (B) 11. $[1.32]$ 12. (A) (D) 13. (A) 14. (D)
 15. (C) 16. $[1 : 49]$